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Huang Qin decoction supplemented diet attenuates *Aeromonas hydrophila*-induced intestinal inflammation *via* the Delta/Notch pathway in *Eriocheir sinensis*

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ABSTRACT

Intestinal inflammation is one of the most common diseases affecting fish and crustaceans in intensive aquaculture operations, adversely affecting the healthy development of the aquaculture industry. Despite its high prevalence in aquatic animals, understanding the pathogenesis and developing targeted therapeutic strategies for intestinal inflammation remains challenging. Huang Qin Decoction (HQD) is a traditional Chinese herbal formula renowned for its potent anti-inflammatory properties, is widely utilized in China for treating inflammatory bowel diseases in various animals. However, the protective effects and underlying molecular mechanisms of HQD on intestinal inflammation in aquatic species have not been elucidated. In this study, we established a model of bacterial intestinal inflammation in the Chinese mitten crabs (*Eriocheir sinensis*) by oral injection with *Aeromonas hydrophila*. We demonstrated that a diet supplemented with HQD was effective in alleviating intestinal inflammation, mitigating intestinal damage, and improving the survival among crabs with intestinal inflammation in this model. Mechanistically, HQD-supplemented diet targets the suppression of Delta/Notch signaling overactivation *in vivo*, decreases the expression of inflammatory response-associated factors, modulates the composition of the intestinal microbiota, and ultimately ameliorates *A. hydrophila*-induced intestinal inflammation in crabs. Thus, HQD-supplemented diet serves to prevent and treat intestinal inflammation in aquatic invertebrates by regulating the equilibrium of the Notch signaling pathway. Our findings underscore the multifaceted role of HQD in the treatment of intestinal inflammatory across different animals, highlight the potential of Notch signaling as a therapeutic target for inflammatory diseases, and provide valuable insights into preventing and treating intestinal inflammation specifically within crustaceans in aquaculture.

1. Introduction

The Chinese mitten crab (*Eriocheir sinensis*) is one of the most significant farmed crustaceans, widely cultured in China. However, environmental pollution and intensive aquaculture have severely impacted the crab industry, leading to a range of diseases (Shi et al., 2024). Among these, bacterial intestinal inflammation occurs frequently in crab aquaculture and is one of the foremost diseases leading to high mortality rates, thus causing catastrophic economic losses in the production of both juvenile and adult crabs (Zhou et al., 2022). Research indicates that intestinal inflammation in crustaceans is mainly caused by infection with *Aeromonas hydrophila*, an opportunistic pathogen widely distributed in aquatic environments. This infection triggers inflammatory responses characterized by persistent damage to the intestinal barrier and dysbiosis of the intestinal microbiota (Battison et al., 2008). Although antibiotics and various veterinary drugs are routinely employed in aquaculture to prevent and treat this condition, their use poses significant drawbacks, including the risk of drug residues and the resistance of pathogenic bacteria to antibiotics (Furuichi et al., 2024; Zhang et al., 2022b). Under this scenario, adding safe, effective, environmentally friendly, and convenient supplements to the diets is currently the ideal and alternative treatment strategy for controlling bacterial diseases and promoting sustainability in aquaculture (Fehrmann-Cartes et al., 2019; Yang et al., 2024b).

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Herbs and their extracts are increasingly recognized as promising feed supplements for aquatic animals due to their significant bacteriostatic and anti-inflammatory properties (Citarasu et al., 2006; Li et al., 2022). For instance, the addition of *Astragalus* polysaccharide to the diet has been shown to significantly increase growth, immunity, and disease resistance in *Apostichopus japonicus* (Fan et al., 2013). Furthermore, dietary supplementation with curcumin has been reported to improve intestinal immune enzyme activity and survival rates in *Seriola dumerili* (He et al., 2022). Inclusion of seabuckthorn in the diet enhances the resistance of grass carp to bacterial infection and foodborne enteritis (Liu et al., 2025). A recent study has demonstrated that *Scutellaria baicalensis* Georgi exhibits extensive antibacterial effects against various aquatic pathogenic bacteria, including *A. hydrophila* (Xia et al., 2020), which has aroused great interest within the aquatic animal feed sector (Sun et al., 2023; Xia et al., 2020). However, the efficacy and therapeutic targets of a single herb are limited. In practical applications, Chinese herbal formula compound with one herb as the core and multiple herbs combined is often used to treat diseases (Zheng et al., 2022b). Huang Qin Decoction (HQD), composed of *Scutellaria baicalensis* Georgi, *Glycyrrhiza uralensis* Fisch, *Paeonia lactiflora* Pall and *Ziziphus jujuba* Mill, is a traditional Chinese herbal formula documented in the Chinese medical book "Shang-Han-Lun". It has been widely utilized for the treatment of common intestinal diseases across various animals and has become a cornerstone in therapeutic strategies for intestinal inflammation (Chen et al., 2015; Lam et al., 2010). Nevertheless, the specific molecular regulatory mechanisms and events of HQD in intestinal inflammation remain poorly understood. More importantly, to date, research on dietary HQD has predominantly focused on enteritis in vertebrates, with limited information available regarding its application in treating intestinal inflammation in aquatic invertebrates. Therefore, investigating the pathogenesis of intestinal inflammation in aquatic invertebrates and finding effective prevention methods and therapeutic targets are the main directions for future breakthroughs.

The canonical Notch pathway is an evolutionarily highly conserved cell signaling system that plays a crucial role in intestinal immunity (Fre et al., 2005), and has been identified as a therapeutic target for inflammatory bowel disease in vertebrates (Jing et al., 2023). For example, Gegen Qinlian decoction has been shown to participate in the maintenance of intestinal mucosal homeostasis by targeting the Notch signaling pathway, thereby alleviating the symptoms of ulcerative colitis (Zheng et al., 2022b). Similarly, the Notch pathway is involved in regulating intestinal homeostasis throughout the lifecycle of zebrafish, with its expression closely linked to intestinal microflora activity and chronic intestinal inflammation (Troll et al., 2018). It should be emphasized that the intestinal microbiota has been utilized to elucidate the unique mechanisms by which traditional Chinese medicine addresses intestinal inflammation (Wang et al., 2024). Notch signaling is activated by the binding of the transmembrane receptors Notch proteins to their corresponding ligands, with different ligands producing varying intensities of Notch signaling and downstream effects (Van de Walle et al., 2011). For instance, Notch1 and its ligand Jagged1 are highly expressed in the hyperplastic crypts from inflamed intestines cells of sulfate sodium (DSS)-induced colitis mice, where they are involved in the process of inflammatory bowel disease (Ghorbaninejad et al., 2019). Also, while the deletion of two Notch ligands, Dll1 and Dll4, results in complete silencing of Notch activation within the intestinal epithelium of mice, Dll4 is identified as the major ligand required for maintaining intestinal epithelial homeostasis (Shimizu et al., 2014). Notably, recent studies in *Litopenaeus vannamei* have shown that ligand Delta-activated Notch signaling is involved in antibacterial immunity (Zhou et al., 2021). Our previous studies on crabs also demonstrated that this signaling could be induced and significantly activated by *Vibrio parahaemolyticus*, thereby regulating the expression of genes related to cell proliferation (Li et al., 2024). Taken together, these findings imply that the core components of Notch signaling, including its receptors and ligands, may play an indispensable role in intestinal immunity, positioning this pathway as a potential target for preventing intestinal inflammation in aquatic animals.

Against the backdrop of the frequent occurrence of bacterial intestinal inflammation in crustaceans, the Chinese mitten crab has emerged as an attractive animal model for aquaculture research due to its conserved intestinal morphology and functions (Che et al., 2024; Cheng et al., 2024; Zhou et al., 2022). In this study, we aim to utilize the crab intestinal inflammation model induced by *A. hydrophila* to investigate the protective effects of HQD-supplemented diet and elucidate its underlying mechanisms for alleviating bacterial intestinal inflammation in crabs. Our findings strongly suggest that dietary supplementation with HQD modulates the Notch signaling pathway, thereby influencing the expression of inflammation-associated factors and altering the composition of the intestinal microbiota to alleviate intestinal inflammation and damage in crabs. Consequently, our research further underscores the therapeutic potential of Chinese herbal compound in treating inflammatory diseases in aquatic animals and provides novel theoretical and therapeutic targets for the prevention and control of intestinal inflammation in crab aquaculture.

2. Materials and methods

2.1. Animals

Healthy Chinese mitten crabs (*E. sinensis*, approximately 17.2 g) were obtained from an aquatic farm located in Suzhou City, Jiangsu Province, China. These crabs were raised in the laboratory under previously described previously (Li et al., 2024). All animal experiments were performed on randomly selected healthy individuals following approval from the Animal Care and Use Ethics Committee of Henan Normal University.

2.2. Oral injection and sample collection

For oral injection experiments, *A. hydrophila* stored in our laboratory was cultured to exponential phase in Luria Bertani (LB) medium at 28 °C, collected by centrifugation, and resuspended in phosphate-buffered saline (PBS) for colony-forming unit (CFU) assays (Zhao et al., 2023b). A sterile flexible microsyringe was employed to inject 150 μl of the indicated concentration of *A. hydrophila* into crab's oral cavity, while control crabs received an equivalent volume of sterile PBS orally. Intestinal samples from crabs in various treatment groups were collected at specific time points following oral injection for RNA extraction, bacterial counts, morphological assessments, and histological analysis, with samples obtained from at least three crabs per group. For the intestinal colonization assay, *A. hydrophila* was labeled with fluorescein isothiocyanate (FITC) (Abcam, UK) (Zhou et al., 2022). Briefly, the collected *A. hydrophila* was heat-inactivated, washed with $NaHCO_3$, and incubated with FITC for 1 h. The FITC-labeled *A. hydrophila* was then washed three times with PBS and resuspended in PBS for CFU assay. After 6 h of oral injection using the same method, the intestinal contents of crabs were collected and imaged under a fluorescence microscope.

2.3. Survival rate assessment and tissue morphology detection

For survival rate analysis, healthy crabs were randomly divided into groups of 30 or 40 individuals, with different groups subjected to varying treatments based on experimental requirements. The number of dead crabs in each group was recorded daily over a period of ten days, and survival curves were statistically analyzed using Kaplan-Meier plots. For histological analysis, the crabs were carefully dissected to obtain their entire intestines which were subsequently washed three times with PBS and photographed. For morphological assessment, the midgut of the crab intestine was utilized. Paraffin sectioning followed by hematoxylin-eosin staining of intestinal tissues were performed according to standard procedures (Zhou et al., 2022), and the sections were examined using a light microscope.

2.4. Gene expression profile analysis

Quantitative real-time PCR (qRT-PCR) was performed to determine the gene expression profiles in the crab intestine. Total RNA was extracted using TRIpure reagent (Aidlab, China), and first-strand cDNA was synthesized using a Reverse Transcriptase kit for qRT-PCR (Takara, Japan), according to the manufacturer's instructions. qRT-PCR was performed using the prepared cDNA, SYBR Green Master Mix (Vazyme, China), and the specific primers listed in Table 1, using the CFX96 Real-Time System (Bio-Rad Laboratories). The cycling procedure comprised 95 °C for 5 min, followed by 40 cycles of 95 °C for 10 s and 60 °C for 30 s, followed melting from 65 °C to 95 °C. Gene expression levels were derived by the $2^{-\Delta\Delta CT}$ method.

2.5. Diet formulation

The preparation of Huang Qin Decoction (HQD) was conducted according to the original prescription outlined in the Chinese Pharmacopoeia (2020). Briefly, *Scutellaria baicalensis* Georgi (9 g), *Paeonia lactiflora* Pall (6 g), *Glycyrrhiza uralensis* Fisch (6 g), and *Ziziphus jujuba* Mill (6 g), purchased from the Pharmacy Department of Henan Normal University Hospital, were mixed, added with 10 times their volume of water, boiled three times, and concentrated to yield a thick decoction with a concentration of 1 g/ml (HQD—H). Subsequently, the thick decoction was diluted tenfold with distilled water to obtain a low concentration HQD at a concentration of 0.1 g/ml (HQD-L).

The preparation of feed pellets coated with HQD (HQD-supplemented diet) was performed as described elsewhere (Dong et al., 2018). Briefly, 1 ml of HQD-H was mixed with 10 g of antibiotic-free commercial feed, and the feed pellets were air-dried for 1 h. Whereas, the control diet (PBS-supplemented diet) was prepared by mixing 1 ml of sterile PBS with 10 g of antibiotic-free commercial feed and subjected to the same drying process. Both diets were stored at 4 °C until further use; in this study, antibiotic-free commercial feed served as the basal diet.

2.6. Antibacterial activity assay

The antimicrobial activity of HQD against *A. hydrophila* was assessed using the Oxford cup method. Briefly, *A. hydrophila* was cultured to its exponential phase in LB medium at a temperature of 28 °C before being mixed with unsolidified LB medium containing agar and poured into plates. Sterile Oxford cups were affixed to these agar plates using tweezers. After the medium had solidified, the Oxford cups were removed, and 200 μl of various concentrations of HQD were added to the wells. Additionally, 200 μl of H_2O and 200 μl of PBS served as negative control, while 200 μl of AMP was utilized as a positive control. The plates were subsequently cultured in a biochemical incubator at 28 °C until zones of inhibition became apparent. The inhibition zone diameters of different groups were measured using a vernier caliper, with all procedures conducted in a bacteria-free environment.

2.7. Bacterial counts in intestine

Crabs from the various treatment groups were carefully dissected as described above to obtain the entire intestines, which were immediately homogenized and diluted in sterile PBS. The diluted homogenate was then dropped and spread onto solid LB medium plates and cultured overnight at 28 °C. Bacterial counts were determined using Image J software to quantify the abundance of culturable microbiota. Each sample was derived from a single crab, with at least five replicates per group.

2.8. RNA interference assay

To knock down the expression levels of *EsNotch* and *EsDelta* in the intestine, partial cDNA fragments of the *EsNotch* and *EsDelta* genes, as well as that for green fluorescent protein (GFP, *Aspergillus constructus*) gene, were amplified using specific primers linked to T7 promoter sequences. Subsequently, double-stranded RNAs (dsRNAs) for the

Table 1
Primers used in this study.*

Primer name	Primer sequence (5′-3′)	Application	Efficiency
qIL-16-F	ACTGTGGTGAGAGTGGTGCTAGG	qRT-PCR	101 %
qIL-16-R	ACGCTTGTGTAAGACGGTGTAACG	qRT-PCR	
qTNF-F	AAGCCACATACACAGTCATCTCTCC	qRT-PCR	98 %
qTNF-R	ATCTCCATCAGCCGTTCCTCC	qRT-PCR	
qLITAF-F	CAGGAGTAGTGTCGGGATTTGC	qRT-PCR	96 %
qLITAF-R	AGTTGTTGGAGCAGCACCTTG	qRT-PCR	
qILF-F	CCGATGGAGACGACCGATAAC	qRT-PCR	101 %
qILF-R	GCACTGTGTGGCACTACTTGC	qRT-PCR	
qP38-MAPK-F	CACTCATGGGTGCTGACCTC	qRT-PCR	97 %
qP38-MAPK-R	TACTTGAGGCCTCGCAACAC	qRT-PCR	
qNotch-F	TCTTGGAACGGTGACTGACATGG	qRT-PCR	98 %
qNotch-R	AGACGATGCCGACGATGACATAG	qRT-PCR	
qDelta-F	AGGGAACCTTCTCGCTCATTATCG	qRT-PCR	100 %
qDelta-R	GTGCTGTGGTCGTCCTCTGTC	qRT-PCR	
qβ-actin-F	GCATCCACGAGACCACTTACA	qRT-PCR	102 %
qβ-actin-R	CTCCTGCTTGCTGATCCACATC	qRT-PCR	
Notch-F	AGACCAAGAACCGCTACA	PCR	
Notch-R	ACTCATTGACATCCCCCT	PCR	
Delta-F	CGGGCGTGAGAGGGTGTG	PCR	
Delta-R	CGGGTGATGAGGGTGGGA	PCR	
GFP-F	TCTGTCAGTGGAGAGGGTGAA	PCR	
GFP-R	TGGTAAAAGGACAGGGCCATC	PCR	
Notch-Ri-F	TAATACGACTCACTATAGGGAGACCAAGAACCGCTACA	Notch dsRNA synthesis	
Notch-Ri-R	TAATACGACTCACTATAGGGACTCATTGACATCCCCCT	Notch dsRNA synthesis	
Delta-Ri-F	TAATACGACTCACTATAGGGCGGGCGTGAGAGGGTGTG	Delta dsRNA synthesis	
Delta-Ri-R	TAATACGACTCACTATAGGGCGGGTGATGAGGGTGGGA	Delta dsRNA synthesis	
GFP-Ri-F	TAATACGACTCACTATAGGGTCTGTCAGTGGAGAGGGTGAA	GFP dsRNA synthesis	
GFP-Ri-R	TAATACGACTCACTATAGGGTGGTAAAAGGACAGGGCCATC	GFP dsRNA synthesis	

* The sequence of the T7 promoter is underlined.

different target genes were synthesized according to the manufacturer's instructions (T7 Transcription Kit, Thermo Fisher Scientific, USA). The synthesized dsRNA was purified and quantified to achieve a final concentration of 2 μg/μl using agarose gel and sterile PBS. Different dsRNA fragments (2 μg/g body weight) were injected into distinct groups of crabs using a microsyringe, with a second injection was carried out 24 h later at an identical dosage. RNA interference efficiency was assessed *via* qRT-PCR at 48 h post-dsRNA administration.

2.9. Statistical analysis

Statistical analyses and graphing were performed using SPSS 24.0 and GraphPad Prism software 8.0. Significance was assessed using Student's *t*-test, with asterisks in figures indicate statistical significance (*$p < 0.05$ and **$p < 0.01$). Detailed statistical information is provided in the corresponding figures and figure legends. All images presented in this study are representative of at least three independent experiments, and all bar charts show the mean ± standard deviation (SD) from a minimum of three replicates.

3. Results

3.1. *A. hydrophila* induces intestinal inflammation in *E. sinensis*

To establish a model of *A. hydrophila*-induced intestinal inflammation in *E. sinensis*, we used oral injection of bacteria to emulate oral infections in nature. We first explored the appropriate concentration and dosage of *A. hydrophila* for our experiments. Ultimately, we selected a dose of 5×10^7 CFU in a volume of 150 μl, as this dosage resulted in a gradual and moderate mortality rate (30–40 %) over a period of 10 days (Fig. 1A). To confirm that orally injected *A. hydrophila* could be effectively delivered and colonized within the crab intestines, we administered FITC-labeled *A. hydrophila* into the oral cavity of crabs. After 6 h, fluorescence microscopy revealed the presence of FITC-labeled *A. hydrophila* within crab intestinal epithelial cells (Fig. 1B). Subsequently, the crabs received daily oral injections with 150 μl of *A. hydrophila* (5×10^7 CFUs per crab) or an equivalent volume of PBS for ten days (Fig. 1B). On the 10th day, the intestinal morphology of the *A. hydrophila* group showed significant intestinal damage with profound oedema compared to the PBS group. Histological analysis further demonstrated alterations such as thinning of both intestinal plica and mucosal layer, atrophy of intestinal villus, crypt damage, detachment of epithelial cells from the basement membrane, and infiltration of numerous inflammatory cells into the mucosa (Fig. 1C). Additionally, the levels of inflammatory response-associated factors *IL-16*, *TNF*, *LITAF*, *ILF*, and *P38-MAPK* were remarkably increased in the *A. hydrophila* group (Fig. 1D). These results align with the classic symptoms of intestinal inflammation in aquatic animals (Pan et al., 2022; Yang et al., 2024a). Thus, these data indicate that we have successfully established the *A. hydrophila*-induced intestinal inflammation model in *E. sinensis*.

3.2. HQD exerts an inhibitory effect on *A. hydrophila*

Next, we aimed to determine whether HQD possesses bacteriostatic activity against *A. hydrophila*, which is responsible for intestinal inflammation in aquatic animals. Bacteriostatic assays conducted using the Oxford cup method demonstrated that high concentrations of HQD, similar to the positive control AMP, exhibited significant antibacterial activity against *A. hydrophila*, as evidenced by clear zones of inhibition

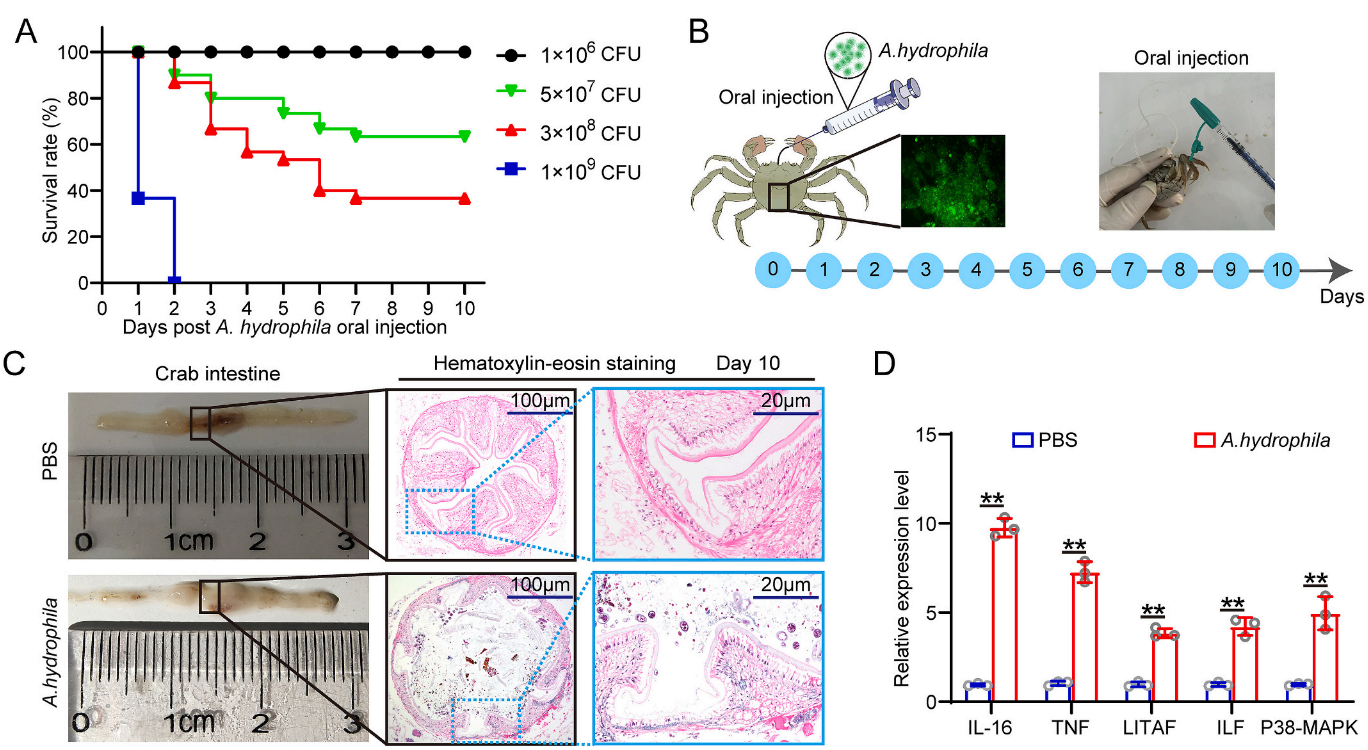

Fig. 1. Establishment of an intestinal inflammation model in *E. sinensis*. (A) Survival percentage of crabs after oral injection with 150 μl of the indicated concentration of *A. hydrophila* (30 crabs per group). (B) Schematic diagram depicting the experimental design for establishing the intestinal inflammation. *A. hydrophila* (150 μl, 5×10^7 CFUs per crab) was administered into the oral cavity of crabs using a sterile flexible microsyringe, daily for a total duration of ten days. (C) Morphological and histological analysis of the intestine on day 10 during the establishment of the intestinal inflammation model in crabs. Intestines from moribund crabs were sampled at specific time points, and images were captured. Tissues were sectioned and stained with hematoxylin-eosin. Data are representative of three independent experiments. Scale bar, 100 or 20 μm. (D) Expression levels of inflammatory response-associated factors in the intestines of crabs on day 10 of the intestinal inflammation model was detected using qRT-PCR. Bars indicate mean ± SD derived from three independent samples, each comprising at least three crabs per sample. Asterisks (*) above bars indicate statistically significant differences (** $p < 0.01$, Student's *t*-test).

observed (Fig. 2A). In contrast, negative controls (H_2O and PBS) exhibited neither bactericidal nor bacteriostatic effects (Fig. 2A). These findings were consistently validated across multiple experimental repetitions (Fig. 2B). Collectively, we conclude that HQD demonstrates effective antimicrobial activity against *A. hydrophila*.

3.3. HQD-supplemented diet promotes recovery of A. hydrophila-induced intestinal inflammation in crabs

Given the remarkable antibacterial capacity of HQD against *A. hydrophila* and its widespread use in treating intestinal diseases in various animals, we sought to explore its therapeutic effect and potential regulatory mechanisms on intestinal inflammation in aquatic animals. We hypothesized that dietary supplementation with HQD could ameliorate intestinal inflammation in *E. sinensis*. To verify this, we treated crabs with intestinal inflammation using either HQD- or PBS-supplemented diets for ten days (Fig. 3A). Initially, we assessed the morality rate of crabs with intestinal inflammation during this ten-day dietary treatment, as lifespan is a critical indicator of intestinal homeostasis regulation in crabs (Zhou et al., 2022). We identified HQD as a promising treatment candidate, as dietary supplementation with HQD significantly improved the survival rates of crabs with intestinal inflammation (Fig. 3B). Subsequent evaluations of intestinal morphology and histology revealed that while improvements were observed in both groups after ten days of treatment, the intestines of crabs in the PBS treatment group remained red and swollen, exhibiting reduced villus height and thinning of the intestinal plica and mucosal layer (Fig. 3C). In contrast, structural disruptions and disturbance of the intestinal crypt and villi were notably ameliorated, and the degree of inflammatory infiltration was significantly reduced in the HQD treatment group (Fig. 3C). These results demonstrate that a short-term oral HQD-supplemented diet offers substantial protection against *A. hydrophila*-induced intestinal inflammation in *E. sinensis*.

3.4. HQD-supplemented diet significantly reduces the expression of inflammatory response-associated factors and the abundance of culturable bacteria in A. hydrophila-induced intestinal inflammation model

Inflammatory response-associated factors and intestinal microbiota play an indispensable role in the development and progression of intestinal inflammation. To further investigate the protective effects of HQD on intestinal inflammation, we examined the expression levels of inflammatory response-associated factors as well as the abundance of culturable bacteria in the intestines. qRT-PCR results indicated that, by day 10 of treatment, the expression levels of inflammatory response-associated factors *IL-16*, *TNF*, *LITAF*, *ILF*, and *P38-MAPK* in the intestine was significantly lower in the HQD treatment group compared to the PBS treatment group (Fig. 4A). Additionally, assessment of intestinal bacterial abundance revealed a significant reduction in the number of culturable intestinal bacteria in the HQD treatment group on the 10th day of treatment (Fig. 4B and C). Based on these findings, it can be concluded that HQD-supplemented diets may exert a protective effect against intestinal inflammation in crabs by downregulating the expression of inflammation-associated factors and modulating the intestinal microbiota composition.

3.5. Involvement of the notch signaling pathway in A. hydrophila-induced intestinal inflammation in crabs

The Notch signaling pathway has been extensively studied for its role in vertebrate intestinal immunity and is recognized as a therapeutic target for inflammatory bowel disease (Jing et al., 2023). However, the regulatory role of the Notch signaling pathway in intestinal inflammation in aquatic animals, and its involvement in HQD-based treatment of intestinal inflammation, remains unclear. To fill this gap, we first examined the expression levels of key receptor Notch and ligand Delta genes within this signaling pathway during the development of intestinal inflammation using qRT-PCR. Our results indicated that, compared to the control group, the expression of both the receptor Notch (Fig. 5A and C) and the ligand Delta (Fig. 5B and C) was rapidly induced during the early phase (approximately 2–3 days post oral injection) of intestinal inflammation and remained highly activated throughout both middle and late stage (approximately 4–10 days post oral injection). These findings suggest that the Delta/Notch pathway is involved in the pathological processes of intestinal inflammation in *E. sinensis*.

3.6. HQD-supplemented diet exerts intestinal immunomodulatory effects via the Delta/notch pathway

In light of the aforementioned results and the known important role of Notch signaling in intestinal immunity (Fre et al., 2005), we next aimed to explore the preliminary effects of the HQD-supplemented diet on this signaling pathway. qRT-PCR analysis revealed that the expression levels of both receptor Notch and ligand Delta genes in the HQD treatment group were significantly lower than those in the PBS treatment group on the 10th day following treatment of crabs with intestinal inflammation (Fig. 6A and B). This preliminary finding gives us reason to hypothesize that the HQD-supplemented diet may alleviate

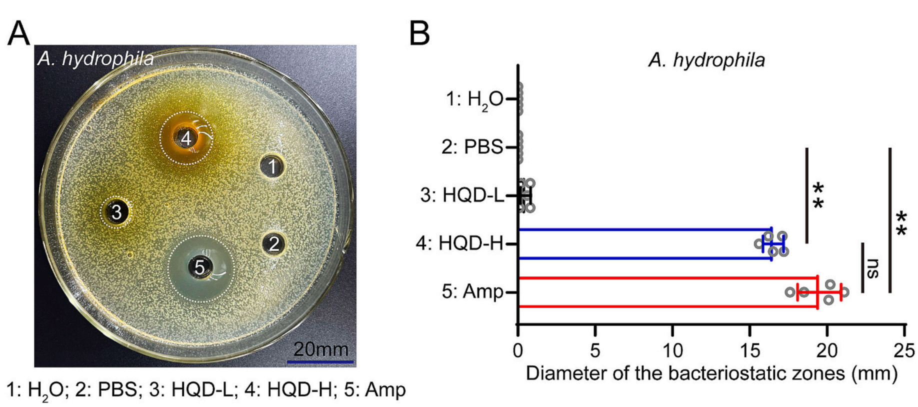

Fig. 2. Antibacterial activity of HQD against *A. hydrophila*. (A) Representative photographs of inhibiting effect of HQD on *A. hydrophila*. *A. hydrophila* was cultured and plated onto LB agar plates. Utilizing the Oxford cup method, 200 μl of H_2O, PBS, low concentration of HQD (0.1 g/ml, designated as HQD-L), high concentration of HQD (1 g/ml, designated as HQD-H) and ampicillin (Amp, 500 μg/ml) were added to wells 1 through 5 respectively. The plates were incubated in a biochemical incubator at a temperature of 28 °C. (B) Bacteriostatic zones for various samples against *A. hydrophila*. The inhibition zone ranges of different groups were measured using a vernier caliper. Data are represented as mean ± SD. Asterisks (*) above bars indicate statistically significant differences (** $p < 0.01$, ns: no significance, Student's *t*-test, $n = 5$).

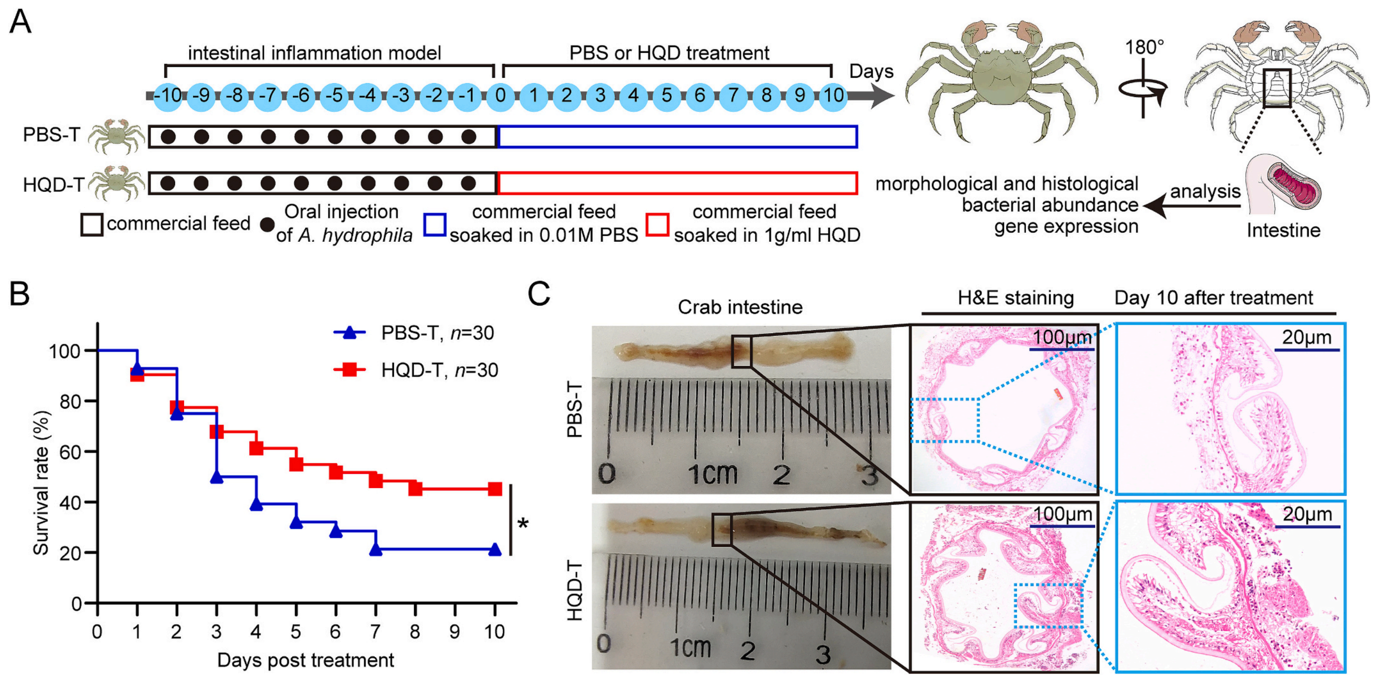

Fig. 3. Protect crab hosts intestine against bacteria-induced inflammation through HQD-supplemented diet. (A) Schematic representation illustrating the dietary treatment process in the *A. hydrophila*-induced intestinal inflammation model in crabs. PBS-supplementation diet treatment group (PBS-T): crabs with intestinal inflammation were fed PBS-coated pellets for ten days; HQD-supplementation diet treatment group (HQD-T): crabs with intestinal inflammation were fed HQD-coated pellets for ten days. (B) HQD-supplemented diet increases the survival rate of crabs with *A. hydrophila*- induced intestinal inflammation. Crabs with intestinal inflammation were randomly allocated to two dietary treatment groups (30 crabs per group) and fed with HQD-supplemented diet (HQD-T) or PBS supplemented diet (PBS-T, control) feed pellets over a period of ten days. Mortality rates were recorded daily throughout this 10-day dietary treatment period. Asterisks (*) above bars indicate statistically significant differences (*$p < 0.05$, Student's *t*-test). (C) Intestinal morphological and histological analysis on day 10 post-treatment with different diets in crabs. Intestines from moribund crabs were sampled at specific time points, and images were captured. Tissues were sectioned and stained with hematoxylin-eosin. Data are representative of three independent experiments. Scale bar, 100 or 20 μm.

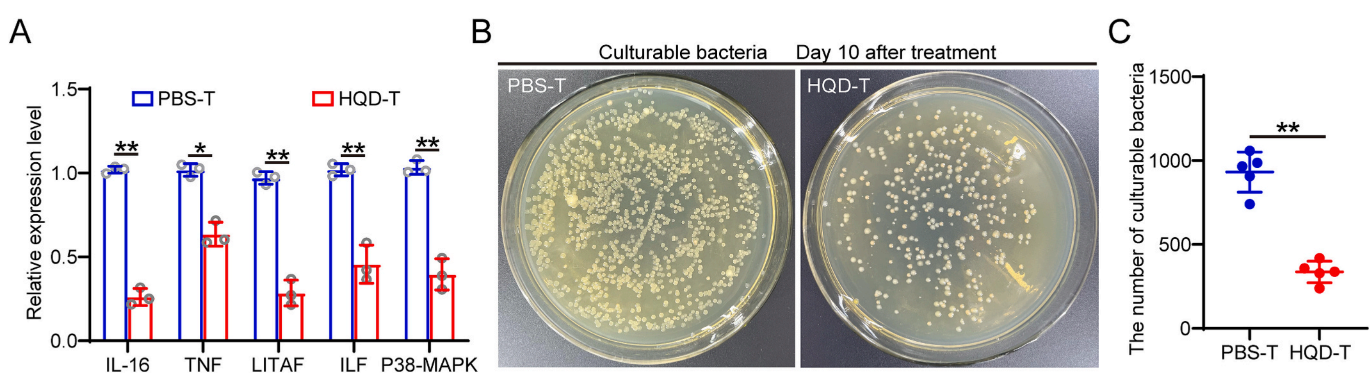

Fig. 4. Effects of HQD-supplemented diet on the expression of inflammatory response-associated factor genes and the abundance of intestinal bacteria. (A) Expression levels of inflammatory response-associated factors in the intestines of crabs on day 10 of dietary treatment was detected using qRT-PCR. Bars indicate mean ± SD derived from three independent samples, each comprising at least three crabs per sample. Asterisks (*) above bars indicate statistically significant differences (** $p < 0.01$, Student's *t*-test). (B) Representative photographs of different dietary treatment affecting the abundance of culturable intestinal bacteria in crabs. Intestinal samples collected on day 10 after dietary treatment in different groups were homogenized, and bacterial load was determined using the plate-counting method at a dilution ratio of 1:10. (C) Influence of different dietary treatments on the total number of culturable intestinal bacteria in crabs. Intestinal homogenates were extracted and cultured on the LB plates for bacterial counts. Each dot represents an individual crab. Data are representative of five independent repeats (** $p < 0.01$, Student's *t*-test).

A. hydrophila-induced intestinal inflammation by suppressing the over-activation of the Notch pathway. To test this hypothesis, we performed RNA interference (RNAi) experiments wherein specific dsRNA administration significantly suppressed the expression levels of *Notch* (Fig. 6C) and *Delta* (Fig. 6D) in the intestines of crabs with intestinal inflammation. Subsequently, we observed that the effect of HQD treatment on the expression of inflammatory response-associated factors was diminished in Notch-knockdown or Delta-knockdown crabs (Fig. 6E and F). That is, knockdown of Notch counteracted the reduction in mRNA expression levels of *IL-6*, *TNF*, *ILF*, and *P38-MAPK* genes caused by HQD treatment (Fig. 6E), and the decline in mRNA expression of *IL-6*, *TNF* and *P38-MAPK* due to HQD treatment was abrogated by Delta silencing (Fig. 6F). Similarly, the regulatory effect of HQD on the abundance and number of culturable bacteria in the intestine was also diminished (Fig. 6G-J). Collectively, these data suggest that the protective effect of the HQD-supplemented diet against intestinal inflammation in crabs may be

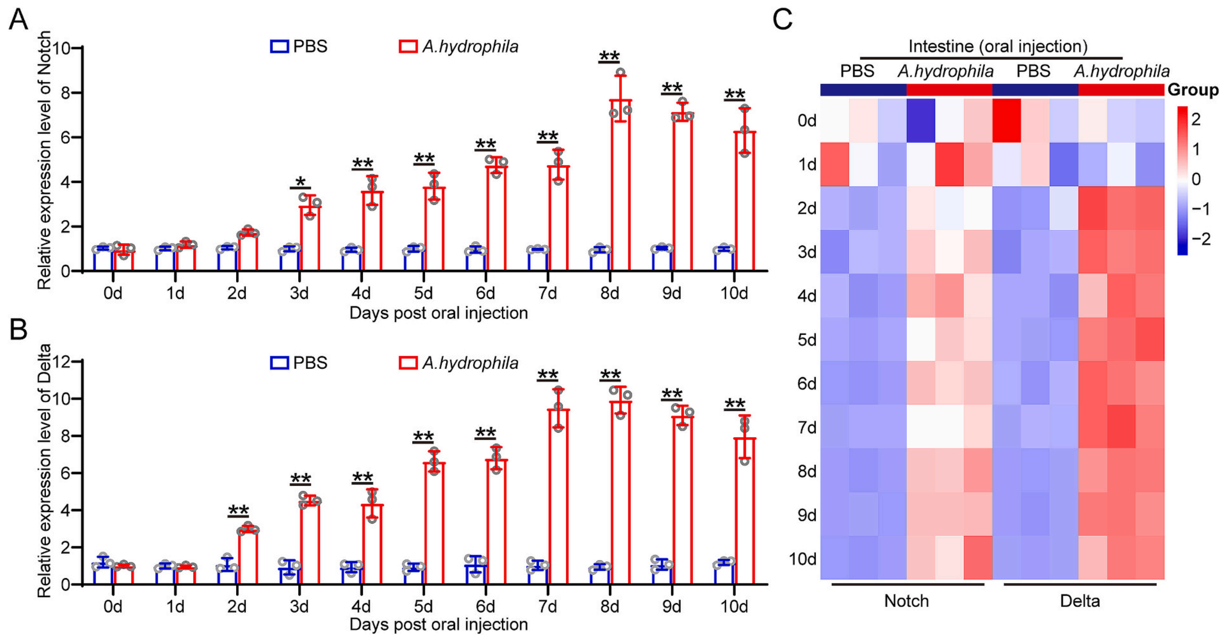

Fig. 5. Involvement of the Delta/Notch pathway in *A. hydrophila*-induced intestinal inflammation in crabs. (A and B) Expression patterns of *Es*Notch (A) and *Es*Delta (B) in the intestines during different phases of intestinal inflammation. Bars indicate mean ± SD derived from three independent samples, each comprising at least three crabs per sample. Asterisks (*) above bars indicate statistically significant differences (*$p < 0.05$, ** $p < 0.01$, Student's *t*-test). (C) mRNA levels of *Es*Notch and *Es*Delta at different phases of intestinal inflammation were converted to a 2-based log function and subsequently normalized by the row using a normalization method. Color scale and Log2 values are shown on the right side of the heatmap, with elevated expression level is indicated by a gradient color change from blue to red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

achieved through its influence on the expression of the Notch signaling pathway.

3.7. HQD-supplemented diet exhibits no protective effects in EsNotch and EsDelta knockdown crabs

We next sought to elucidate whether Notch signaling serves as an effector pathway for HQD treatment of intestinal inflammation in crabs. On the 10th day after HQD treatment of Notch-knockdown or Delta-knockdown crabs, we performed histological analysis and staining of moribund crab intestines to assess any histological or morphological changes. As expected, the intestines from both Notch- or Delta-silenced crabs exhibited significant morphological alterations and tissue damage despite treatment with the HQD-supplemented diet for ten days (Fig. 7A). Specific observations included thinning of the intestinal plica and mucosal layer, damage to intestinal tissue integrity and crypts, as well as absence of feces in the intestine. Additionally, severe inflammatory cell infiltration was clearly observed in the Notch-silenced group, while notable mucosal injury was evident in the Delta-silenced group (Fig. 7A). These observations suggest that Notch signaling plays an important role in the HQD treatment of *A. hydrophila*-induced intestinal inflammation. To further determine the mechanism, we ultimately also analyzed the survival rates of crabs subjected to the HQD-supplemented diet in the Notch-knockdown or Delta-knockdown groups over a 10-day treatment period. The results revealed that both knockdown groups exhibited similar survival rates following HQD treatment, both significantly lower than the control group. Specifically, more than 40 % of the crabs in the control group remained alive on the 10th day post-treatment, whereas the survival rate of both experimental groups fell below 20 % (Fig. 7B). Taken together, these results support the conclusion that the HQD-supplemented diet protects crabs from intestinal inflammation through the Notch signaling pathway.

4. Discussion

For the currently rapidly growing aquaculture industry, the biggest risks comes from various diseases (Lee et al., 2022). Among them, the top-ranked diseases include intestinal inflammation, although this disease is always underestimated (Cheng et al., 2024). In this study, we established a bacterial intestinal inflammation model in crabs through oral injection of *A. hydrophila*, and demonstrated that HQD-supplemented diet may alleviate this inflammation by suppressing Notch signaling-mediated inflammation-associated factors expression and intestinal microbiota homeostasis regulation. These findings provide novel insights into the occurrence and development of intestinal inflammation in aquatic animals, as well as potential therapeutic strategies.

The intestinal inflammation model is considered to be an indispensable basic tool for studying intestinal inflammation in animals (Cheng et al., 2024; Yan et al., 2024; Zhou et al., 2022), making it imperative to establish an animal model that accurately mirrors bacterial intestinal inflammation in crustaceans. In the natural environment, the route of infection for bacterial intestinal inflammation is mainly through oral ingestion. Therefore, oral delivery of pathogens more closely mimics the natural infection route, as opposed to injections (Wang et al., 2014). The intestine tissues of crustaceans, including the Chinese mitten crab, are the targeted tissue after bacterial oral infection and are among the organs most susceptible to inflammation and lesions (Che et al., 2024; Zhou et al., 2022). However, one of the challenges in establishing an intestinal inflammation model in the laboratory is determining the appropriate bacterial dosages. *A. hydrophila* has been shown to induce intestinal flora disturbances (Ci et al., 2024), morphological damage (Jin et al., 2023) and barrier dysfunction (Wu et al., 2024), which ultimately leads to bacterial intestinal inflammation in aquatic animals. Research on *Cyprinus carpio* indicated that intestinal immunity was significantly activated after infection with 4.87×10^7 CFU/ml of *A. hydrophila* (Chen et al., 2020). In our study, we used a dosage of 5×10^7 CFUs per crab for the infection with *A. hydrophila*. This

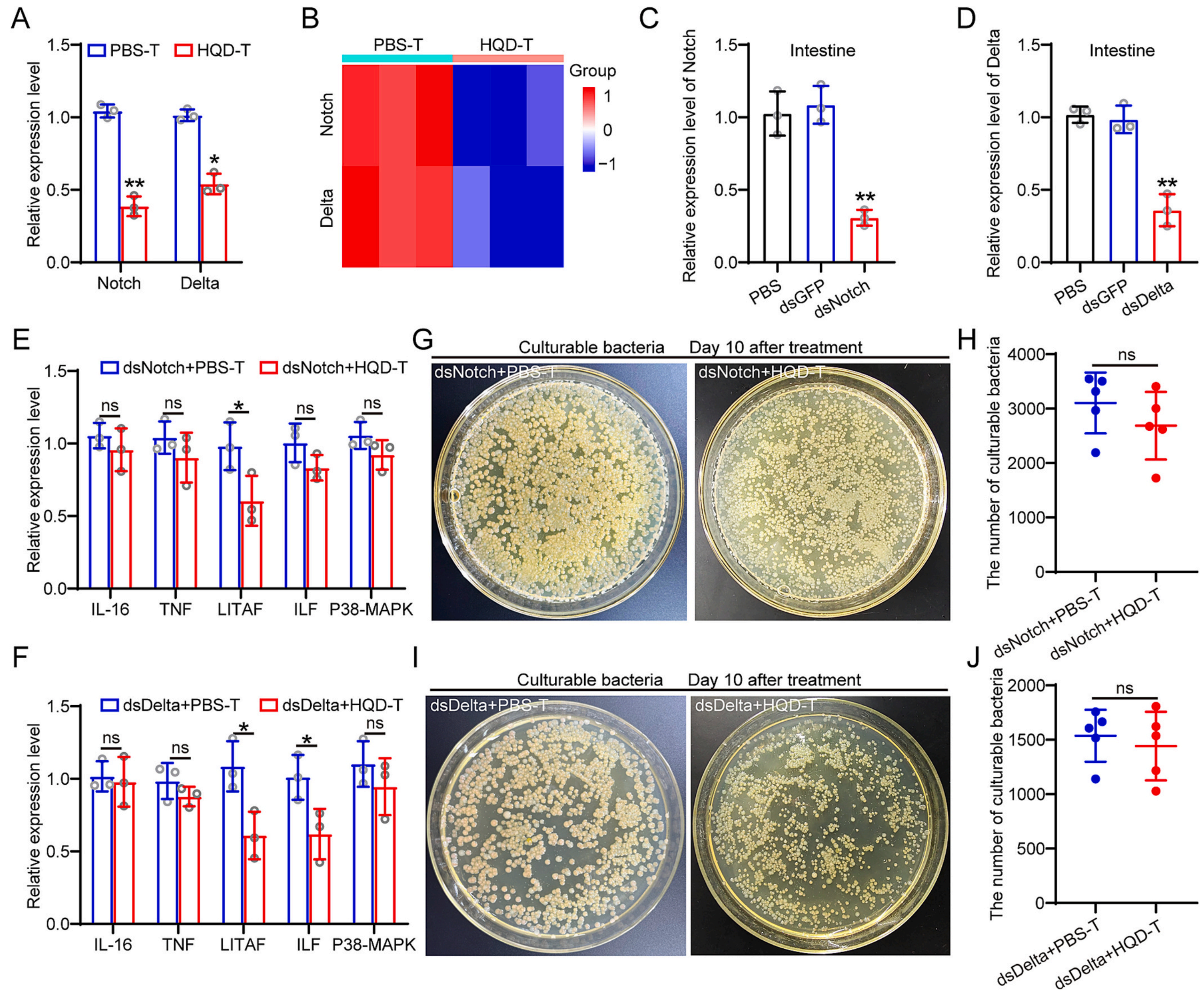

Fig. 6. HQD-supplemented diet exerts immunomodulatory effects on the intestine *via* the Delta/Notch pathway. (A) Expression levels of *Es*Notch and *Es*Delta in the intestines of crabs on day 10 of dietary treatment were detected using qRT-PCR. Bars indicate mean ± SD derived from three independent samples, each comprising at least three crabs per sample. Asterisks (*) above bars indicate statistically significant differences ($*p < 0.05$, $** p < 0.01$, Student's *t*-test). (B) mRNA levels of *Es*Notch and *Es*Delta in the intestines of crabs on day 10 of dietary treatment were converted to a 2-based log function and subsequently normalized by the row using a normalization method. Color scale and Log2 values are shown on the right side of the heatmap, with elevated expression level is indicated by a gradient color change from blue to red. (C and D) RNAi efficiency of *Es*Notch (C) and *Es*Delta (D) in crab intestines was evaluated using qRT-PCR. dsRNA was injected into crabs at a dosage of 2 μg/g body weight. mRNA expression levels of *Es*Notch and *Es*Delta in the intestine were detected at specified time points. Results are expressed as means ± SD from three independent experiments ($** p < 0.01$, Student's *t*-test). (E and F) Effects of dietary treatment on the expression of inflammatory response-associated factors in the intestine after silencing *Es*Notch (E) and *Es*Delta (F). Bars indicate mean ± SD derived from three independent samples, each comprising at least three crabs per sample. Asterisks (*) above bars indicate statistically significant differences ($*p < 0.05$, $** p < 0.01$, Student's *t*-test). (G and I) Representative photographs of different dietary treatment affecting the abundance of culturable intestinal bacteria in crabs after silencing *Es*Notch (G) and *Es*Delta (I). Intestines tissues from *Es*Notch-knockdown or *Es*Delta-knockdown crabs were homogenized on the 10th day after dietary treatment; bacterial loads were determined using the plate-counting method at a dilution ratio of 1:10. (H and J) Influence of different dietary treatments on the total number of culturable intestinal bacteria in crabs after silencing *Es*Notch (H) and *Es*Delta (J). Intestinal homogenates were extracted and cultured on LB plates for bacterial counts. Each dot represents an individual crab. Data are representative of five independent repeats (ns: no significance, Student's *t*-test). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

dosage was chosen because, on the one hand, it ensured that enough crabs survived for sampling, and on the other hand, this dosage also induced intestinal inflammation in surviving crabs, which was confirmed by subsequent inflammatory response-associated factors expression and intestinal histopathological observation. The inflammatory cytokines IL-16 and TNF, along with the inflammatory transcription factors LITAF and ILF, and the important related-factor of the inflammatory response P38-MAPK, are essential for detecting inflammatory response (Zhao et al., 2023a). Additionally, histological changes in the intestine are usually closely associated with intestinal inflammation (Ci et al., 2024; Jin et al., 2023; Wu et al., 2024). Our results demonstrated significant intestinal damage after ten days of continuous challenge with *A. hydrophila*, with marked upregulation of inflammatory response-associated factors (*IL-16*, *TNF*, *LITAF*, *ILF* and *P38-MAPK*) in the intestines (Fig. 1). Similar findings have been reported in models of *A. hydrophila*-induced bacterial intestinal inflammation in crucian carp (Dong et al., 2018), yellow catfish (Ci et al., 2024), snakehead (Wu et al., 2024), and blunt snout bream (Zhang et al., 2022a). These results

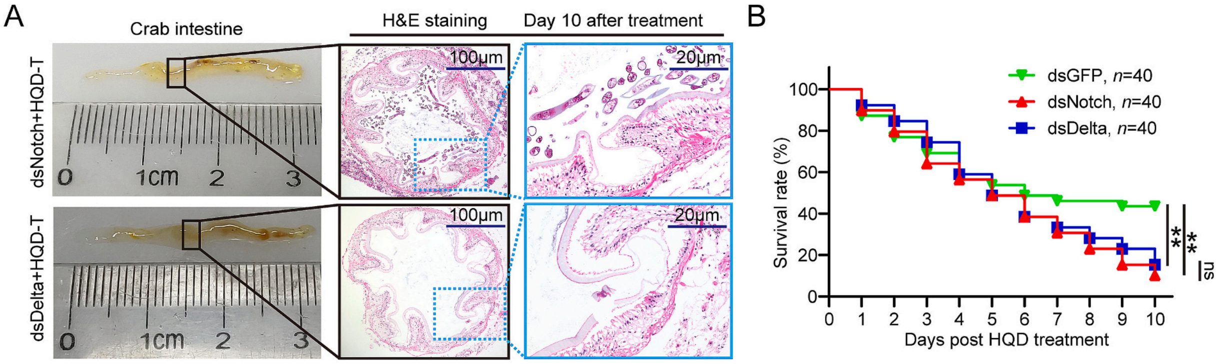

Fig. 7. HQD-supplemented diet exhibits no protective effects in *Es*Notch and *Es*Delta knockdown crabs. (A) Intestinal morphological and histological analysis on the 10th day of treatment with HQD-supplemented diet in *Es*Notch or *Es*Delta knockdown crabs. Intestines from moribund crabs were sampled at specific time points, and images were captured. Tissues were sectioned and stained with hematoxylin-eosin. Data are representative of three independent experiments. Scale bar, 100 or 20 μm. (B) Silencing *Es*Notch or *Es*Delta diminishes the protective effect of HQD-supplemented diets on crabs with intestinal inflammation. Crabs with intestinal inflammation were injected with dsGFP, ds*Es*Notch, or ds*Es*Delta (40 crabs per group), and subsequently fed HQD-coated (HQD-T) feed pellets for ten days. Mortality was recorded daily throughout the 10-day dietary treatment period. Asterisks (*) above bars indicate statistically significant differences (*$p < 0.05$, Student's *t*-test).

indicate that *A. hydrophila* induces the occurrence of intestinal damage and inflammation in crabs. Importantly, this crab model enhances our understanding of the pathology of intestinal inflammation in crustaceans and provides a tool and platform for screening potential therapeutic drugs and evaluating their efficacy in treating intestinal inflammation.

The adverse reactions of antibiotics and other drugs, incomplete relief of symptoms and the emergence of pathogen resistance all highlight the important needs of Chinese herbal formula in the prevention and control of aquatic animal diseases. HQD has shown considerable promise in treating various animal enteritis diseases (Chen et al., 2016; Chen et al., 2015; Lam et al., 2010), and increasing evidence supports that its significant antimicrobial effects or comprehensive regulation of signaling pathways are primary mechanisms through which HQD treats intestinal inflammation (Li et al., 2020; Mo et al., 2022). In this study, we first observed that HQD exhibited significant antibacterial activity against *A. hydrophila* (Fig. 2), suggesting its potential as an effective herb-based feed additive for preventing and treating bacterial diseases caused by *A. hydrophila* in aquatic animals. Subsequently, we demonstrated the therapeutic effect of the HQD-supplemented diet on crab intestinal inflammation based on our successfully established crab model. The HQD-supplemented diet reduced mortality in crabs with intestinal inflammation, likely because its ability to alleviate intestinal injury and inflammation (Fig. 3). The intestine serves not only as a critical site for digestion and nutrient absorption but also as a vital immune defense barrier for crabs (Che et al., 2024; Cheng et al., 2024). Therefore, maintaining intestinal health is crucial for the survival of crabs (Zhou et al., 2022). The occurrence and development of intestinal inflammation are particularly susceptible to the expression levels of inflammatory response-associated factors, as excessive or persistent production of these factors can further compromise the intestinal barrier and exacerbate dysbiosis of the intestinal microbiota (Pan et al., 2022; Wu et al., 2024). Our data also demonstrate that HQD-supplemented diet significantly reduces the expression levels of inflammatory factors and the load of culturable bacteria in the intestine after ten days of treatment (Fig. 4). It is well known that high bacterial loads typically indicate uncontrolled proliferation of newly invasive bacteria in the intestine (Zhang et al., 2021). Studies on zebrafish with intestinal inflammation has shown that bacterial counts in multiple tissues, including the intestine, were significantly higher than those in healthy fish (Liu et al., 2014). Consistent with this, existing studies have widely shown that the intestinal microbiota and their components are key regulators mediating host intestinal inflammation while also influencing host susceptibility to disease (Wu et al., 2024; Yan et al., 2024). Therefore, combined with our current results, we can preliminarily conclude that the HQD-supplemented diet reverses *A. hydrophila*-induced intestinal inflammation and may serve as a promising candidate for treating intestinal inflammation in crustaceans.

In addition to its recognized role in promoting cell differentiation and fate determination, canonical Notch signaling has recently demonstrated accompanying benefits in inflammatory responses, which expands its potential as an immunotherapeutic target. In mammals, Notch1 has been shown to play a significant role in the inflammatory response by regulating cytokine expression (Outtz et al., 2010). In the invertebrate L. *vannamei*, the receptor Notch and ligand Delta regulate the expression of inflammation-related genes and reactive oxygen species (ROS) generation, participating in shrimp immune defense (Ning et al., 2018; Zhou et al., 2021). Interestingly, only one Notch receptor has been identified in invertebrates, and this receptor, along with the ligand Delta, exhibits high adhesion in Drosophila (Ahimou et al., 2004), as demonstrated in our previous studies on crabs (Li et al., 2024). Therefore, to characterize these additional roles of Notch signaling, we first examined the expression patterns of the receptor Notch and ligand Delta during the development of intestinal inflammation. Our results indicated that they are involved in the whole process of intestinal inflammation, suggesting activation of this pathway in the pathological context of such conditions (Fig. 5). Furthermore, after ten days of treatment with the HQD-supplemented diet, we observed significantly reduced expression levels of *Notch* and *Delta* in the intestines of crabs with intestinal inflammation. This finding underscores its significant therapeutic effect since overactivation of Notch signaling often triggers more serious diseases including intestinal barrier injury (Shaoyong et al., 2023). The effects of the HQD-supplemented diet on inflammatory factors expression and intestinal microbiota were eliminated after knockdown of *Notch* and *Delta*, similar outcomes were noted for subsequent histopathological analyses as well as endpoint variables such as mortality rates. Namely, we employed dsRNA to knockdown *Notch* and *Delta* and further identified Notch signaling as a downstream effector target for the therapeutic effects of the HQD-supplemented diet. Furthermore, we explored its relationship with the intestinal histopathology of crabs after *A. hydrophila* infection. However, while our findings provide valuable insights, it is crucial to acknowledge the potential limitations of this study. First, the process of intestinal inflammation involves the co-regulation of multiple signaling pathways and modulators (Zhang et al., 2022a; Zheng et al., 2022a). It remains unclear whether other signaling pathways play significant roles in the protective effects of HQD against *A. hydrophila* challenge in the intestine. Therefore, it is too early to deduce the Delta/Notch pathway serves as the main and only target signal for HQD-supplemented diet to alleviate intestinal inflammation. Additionally, mounting evidence suggests a strong correlation between intestinal microbiota and intestinal inflammatory diseases (Ci et al., 2024; Qi et al., 2024), which highlights the necessity for comprehensive analysis of the intestinal microbiota. With these considerations in mind, our future studies will focus on employing multi-omics sequencing in conjunction with molecular biology experiments to further screen potential action targets for intestinal inflammatory therapies and elucidate their interactions with microbes.

In conclusion, we established a crab model of intestinal inflammation induced by the widespread aquatic pathogen *A. hydrophila* and underscored the therapeutic potential of the HQD-supplemented diet in alleviating crab intestinal inflammation through modulation of the Notch pathway. Consequently, these findings not only indicate to some extent the importance of the Notch signaling pathway in the pathogenesis of intestinal inflammation, but also opens up avenues for the development of more effective treatments for intestinal inflammation in aquatic animals.

CRediT authorship contribution statement

Hao Li: Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Weijia Liu:** Validation, Software, Data curation. **Lei Zhu:** Validation, Software, Data curation. **Libo Hou:** Validation, Software, Data curation. **Xianghui Kong:** Writing – original draft, Visualization, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no conflicts of interest with the contents of this article. All authors agree with the submission. This work has not been published or submitted for publication elsewhere, either completely or in part, or in another form or language.

Data availability

Data will be made available on request.

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References

Ahimou, F., Mok, L.P., Bardot, B., Wesley, C., 2004. The adhesion force of notch with Delta and the rate of notch signaling. J. Cell Biol. 167, 1217–1229.

Battison, A.L., Després, B.M., Greenwood, S.J., 2008. Ulcerative enteritis in Homarus americanus: case report and molecular characterization of intestinal aerobic bacteria of apparently healthy lobsters in live storage. J. Invertebr. Pathol. 99, 129–135.

Che, S., Huang, M., Zhu, L., Shen, L., Ma, Y., Wan, Z., Li, X., Zhou, J., Ding, S., Li, X., 2024. Exposure to nanopolystyrene and phoxim at ambient concentrations causes oxidative stress and inflammation in the intestines of the Chinese mitten crab (Eriocheir sinensis). Ecotoxicol. Environ. Saf. 273, 116126.

Chen, G., Yang, Y., Hu, C., Cheng, X., Xu, Y., Cai, X., Wang, M., Yang, C.S., Cao, P., 2016. Protective effects of Huangqin decoction against ulcerative colitis and associated cancer in mice. Oncotarget 7.

Chen, J., Liu, N., Zhang, H., Zhao, Y., Cao, X., 2020. The effects of Aeromonas hydrophila infection on oxidative stress, nonspecific immunity, autophagy, and apoptosis in the common carp. Develop. Comparat. Immunol. 105, 103587.

Chen, P., Zhou, X., Zhang, L., Shan, M., Bao, B., Cao, Y., Kang, A., Ding, A., 2015. Anti-inflammatory effects of Huangqin tang extract in mice on ulcerative colitis. J. Ethnopharmacol. 162, 207–214.

Cheng, Y., Li, C., Chen, Y., Liu, H., Wan, H., Kang, X., Guo, S., 2024. Water pollutant cadmium-induced intestinal inflammation in Eriocheir sinensis and potential therapeutic drugs. Sci. Total Environ. 955, 177132.

Ci, Y., Ku, T., Su, Y., He, Z., Zhang, Y., Ji, J., Ning, X., Yin, S., Zhang, K., 2024. Response signatures of intestinal microbiota and gene transcription of yellow catfish (Pelteobagrus fulvidraco) to Aeromonas hydrophila infection. Fish Shellfish Immunol. 152, 109797.

Citarasu, T., Sivaram, V., Immanuel, G., Rout, N., Murugan, V., 2006. Influence of selected Indian immunostimulant herbs against white spot syndrome virus (WSSV) infection in black tiger shrimp, Penaeus monodon with reference to haematological, biochemical and immunological changes. Fish Shellfish Immunol. 21, 372–384.

Dong, Y., Yang, Y., Liu, J., Awan, F., Lu, C., Liu, Y., 2018. Inhibition of Aeromonas hydrophila-induced intestinal inflammation and mucosal barrier function damage in crucian carp by oral administration of Lactococcus lactis. Fish Shellfish Immunol. 83, 359–367.

Fan, Y., Yu, X.Q., Xu, L., Wang, S.X., Ye, H.B., Diao, J., Yang, X.S., Li, T.B., 2013. Synergy of microcapsule polysaccharides and on the growth, immunity and resistance of sea cucumber against infection. Fisher. Sci. 79, 807–814.

Fehrmann-Cartes, K., Coronado, M., Hernández, A.J., Allende, M.L., Feijoo, C.G., 2019. Anti-inflammatory effects of aloe vera on soy meal-induced intestinal inflammation in zebrafish. Fish Shellfish Immunol. 95, 564–573.

Fre, S., Huyghe, M., Mourikis, P., Robine, S., Louvard, D., Artavanis-Tsakonas, S., 2005. Notch signals control the fate of immature progenitor cells in the intestine. Nature 435, 964–968.

Furuichi, M., Kawaguchi, T., Pust, M.M., Yasuma-Mitobe, K., Plichta, D.R., Hasegawa, N., Ohya, T., Bhattarai, S.K., Sasajima, S., Aoto, Y., et al., 2024. Commensal consortia decolonize Enterobacteriaceae via ecological control. Nature 633 (8031), 878–886.

Ghorbaninejad, M., Heydari, R., Mohammadi, P., Shahrokh, S., Haghazali, M., Khanabadi, B., Meyfour, A., 2019. Contribution of NOTCH signaling pathway along with TNF-α in the intestinal inflammation of ulcerative colitis. Gastroenterol. Hepatol. Bed. Bench 12, S80–s86.

He, Y., Fu, Z., Dai, S., Yu, G., Ma, Z., Wang, X., 2022. Dietary curcumin supplementation enhances intestinal immunity and gill protection in juvenile greater amberjack (Seriola dumerili). Heliyon 8, e11887.

Jin, W., Jiang, L., Hu, S., Zhu, A., 2023. Metabolite features of serum and intestinal microbiota response of largemouth bass (Micropterus salmoides) after Aeromonas hydrophila challenge. Comp. Biochem. Physiol. C Toxicol. Pharmacol. 263, 109496.

Jing, S., Chen, H., Liu, E., Zhang, M., Zeng, F., Shen, H., Fang, Y., Muhitdinov, B., Huang, Y., 2023. Oral pectin/oligochitosan microspheres for colon-specific controlled release of quercetin to treat inflammatory bowel disease. Carbohydr. Polym. 316, 121025.

Lam, W., Bussom, S., Guan, F., Jiang, Z., Zhang, W., Gullen, E.A., Liu, S.H., Cheng, Y.C., 2010. The four-herb Chinese medicine PHY906 reduces chemotherapy-induced gastrointestinal toxicity. Sci. Transl. Med. 2, 45ra59.

Lee, D., Yu, Y.B., Choi, J.H., Jo, A.H., Hong, S.M., Kang, J.C., Kim, J.H., 2022. Viral shrimp diseases listed by the OIE: a review. Viruses 14.

Li, H., Tang, L., Zhu, L., Hou, L., Kong, X., 2024. Specific notch receptor and its ligand Delta interactions control antibacterial immunity by regulating hemocyte proliferation in Eriocheir sinensis. Aquaculture 593, 741358.

Li, M., Zhao, X., Xie, J., Tong, X., Shan, J., Shi, M., Wang, G., Ye, W., Liu, Y., Unger, B.H., et al., 2022. Dietary inclusion of Seabuckthorn (Hippophae rhamnoides) mitigates foodborne enteritis in zebrafish through the gut-liver immune Axis. Front. Physiol. 13, 831226.

Li, M.Y., Luo, H.J., Wu, X., Liu, Y.H., Gan, Y.X., Xu, N., Zhang, Y.M., Zhang, S.H., Zhou, C.L., Su, Z.R., 2020. Anti-inflammatory effects of Huangqin decoction on dextran sulfate sodium-induced ulcerative colitis in mice through regulation of the gut microbiota and suppression of the Ras-PI3K-Akt-HIF-1α and NF-κB pathways. Front. Pharmacol. 10.

Liu, X., Chang, X., Wu, H., Xiao, J., Gao, Y., Zhang, Y., 2014. Role of intestinal inflammation in predisposition of Edwardsiella tarda infection in zebrafish (Danio rerio). Fish Shellfish Immunol. 41, 271–278.

Liu, Y., Su, L., Cao, D., Wu, N., Hu, Y., Altaf, F., Li, J., Zhu, Q., Liu, J., Zhang, Q., et al., 2025. Comparative analysis of Seabuckthorn and its significant component in enhancing fish mucosal immunity. Aquaculture 596, 741752.

Mo, X., Tang, K., Deng, L., Zhou, X., Li, X., Zhang, Y., Wang, J., 2022. Prevention of ulcerative colitis by Huangqin decoction: reducing the intestinal epithelial cell apoptosis rate through the IFN-γ/JAK/ETS signalling pathway. Pharm. Biol. 60, 1116–1125.

Ning, P., Zheng, Z., Aweya, J.J., Yao, D., Li, S., Ma, H., Wang, F., Zhang, Y., 2018. Litopenaeus vannamei notch affects lipopolysaccharides induced reactive oxygen species. Develop. Comparat. Immunol. 81, 74–82.

Outtz, H.H., Wu, J.K., Wang, X., Kitajewski, J., 2010. Notch1 deficiency results in decreased inflammation during wound healing and regulates vascular endothelial growth factor receptor-1 and inflammatory cytokine expression in macrophages. J. Immunol. 185, 4363–4373.

Pan, M., Liu, D., Liu, J., Li, X., Huang, D., Luo, K., Liu, Y., Wu, Z., Zhang, W., Mai, K., 2022. Biotin alleviates hepatic and intestinal inflammation and apoptosis induced by high dietary carbohydrate in juvenile turbot (Scophthalmus maximus L.). Fish Shellfish Immunol. 130, 560–571.

Qi, Q., Zhang, H., Jin, Z., Wang, C., Xia, M., Chen, B., Lv, B., Peres Diaz, L., Li, X., Feng, R., et al., 2024. Hydrogen sulfide produced by the gut microbiota impairs host metabolism via reducing GLP-1 levels in male mice. Nat Metab. 6 (8), 1601–1615.

Shaoyong, W., Jin, H., Jiang, X., Xu, B., Liu, Y., Wang, Y., Jin, M., 2023. Benzo [a] pyrene-loaded aged polystyrene microplastics promote colonic barrier injury via oxidative stress-mediated notch signalling. J. Hazard. Mater. 457, 131820.

Shi, C., Lin, T.H., Qu, C., 2024. The role of pattern recognition receptors in the innate immune system of Chinese mitten crab (Eriocheir sinensis). Fish Shellfish Immunol. 109946.

Shimizu, H., Okamoto, R., Ito, G., Fujii, S., Nakata, T., Suzuki, K., Murano, T., Mizutani, T., Tsuchiya, K., Nakamura, T., et al., 2014. Distinct expression patterns of notch ligands, Dll1 and Dll4, in normal and inflamed mice intestine. PeerJ 2, e370.

Sun, L., Lin, F., Sun, B., Qin, Z., Chen, K., Zhao, L., Li, J., Zhang, Y., Lin, L., 2023. Scutellaria polysaccharide mediates the immunity and antioxidant capacity of giant freshwater prawn (Macrobrachium rosenbergii). Develop. Comparat. Immunol. 143, 104678.

Troll, J.V., Hamilton, M.K., Abel, M.L., Ganz, J., Bates, J.M., Stephens, W.Z., Melancon, E., van der Vaart, M., Meijer, A.H., Distel, M., et al., 2018. Microbiota promote secretory cell determination in the intestinal epithelium by modulating host notch signaling. Development 145.

Van de Walle, I., De Smet, G., Gärtner, M., De Smedt, M., Waegemans, E., Vandekerckhove, B., Leclercq, G., Plum, J., Aster, J.C., Bernstein, I.D., et al., 2011. Jagged2 acts as a Delta-like notch ligand during early hematopoietic cell fate decisions. Blood 117, 4449–4459.

Wang, X.W., Xu, Y.H., Xu, J.D., Zhao, X.F., Wang, J.X., 2014. Collaboration between a soluble C-type lectin and Calreticulin facilitates white spot syndrome virus infection in shrimp. J. Immunol. 193, 2106–2117.

Wang, Y., Li, C., Li, J., Zhang, S., Zhang, Q., Duan, J., Guo, J., 2024. Abelmoschus manihot polysaccharide fortifies intestinal mucus barrier to alleviate intestinal inflammation by modulating Akkermansia muciniphila abundance. Acta Pharm. Sin. B 14, 3901–3915.

Wu, X.Q., Chen, X.M., Pan, Y.Y., Sun, C., Tian, J.X., Qian, A.D., Niu, X.T., Kong, Y.D., Li, M., Wang, G.Q., 2024. Changes of intestinal barrier in the process of intestinal inflammation induced by Aeromonas hydrophila in snakehead (Channa argus). Fish Shellfish Immunol. 152, 109775.

Xia, Y.-T., Chan, G.K.-L., Wang, H.-Y., Dong, T.T.-X., Duan, R., Hu, W.-H., Qin, Q.-W., Wang, W.-X., Tsim, K.W.-K., 2020. The anti-bacterial effects of aerial parts of Scutellaria baicalensis: potential application as an additive in aquaculture feedings. Aquaculture 526, 735418.

Yan, S., Du, R., Yao, W., Zhang, H., Xue, Y., Teligun, Li, Bao, H., Zhao, Y., Cao, S., et al., 2024. Host-microbe interaction-mediated resistance to DSS-induced inflammatory enteritis in sheep. Microbiome 12, 208.

Yang, H., Mo, A., Yi, L., Wang, J., He, X., Yuan, Y., 2024a. Selenium attenuated food borne cadmium-induced intestinal inflammation in red swamp crayfish (Procambarus clarkii) via regulating PI3K/Akt/NF-κB pathway. Chemosphere 349, 140814.

Yang, Z., Li, J., Ma, Y., Wu, Z., Li, J., Wang, F., Xi, Y., Jiang, Y., Huang, S., Yi, Q., 2024b. Effects of dietary bio-fermented selenium supplementation on growth, immune performance, and intestinal microflora of Chinese mitten crabs, Eriocheir sinensis. Int. J. Mol. Sci. 25.

Zhang, C., Yuan, X., Xu, R., Qi, Q., Wang, Y., 2022a. The intestinal histopathology, innate immune response and antioxidant capacity of blunt snout bream (Megalobrama amblycephala) in response to Aeromonas hydrophila. Fish Shellfish Immunol. 124, 525–533.

Zhang, W., Zhao, J., Ma, Y., Li, J., Chen, X., 2022b. The effective components of herbal medicines used for prevention and control of fish diseases. Fish Shellfish Immunol. 126, 73–83.

Zhang, Y.X., Zhang, M.L., Wang, X.W., 2021. C-type lectin maintains the homeostasis of intestinal microbiota and mediates biofilm formation by intestinal Bacteria in shrimp. J. Immunol. 206, 1140–1150.

Zhao, X., Huang, S., Zhang, P., Qiao, X., Liu, Y., Dong, M., Yi, Q., Wang, L., Song, L., 2023a. A circadian clock protein cryptochrome inhibits the expression of inflammatory cytokines in Chinese mitten crab (Eriocheir sinensis). Int. J. Biol. Macromol. 253, 126591.

Zhao, Y., Qiao, D., Zhang, J., Gao, F., Pei, C., Li, C., Kong, X., 2023b. Activation mechanism of CcGSDMEb-1/2 and regulation for bacterial clearance in common carp (Cyprinus carpio). J. Immunol. 211, 658–672.

Zheng, N., Wang, N., Wang, Z.Y., Abdallah, G., Zhang, B.Y., Wang, S., Yao, Q., Chen, Y. K., Wang, Q.J., Zhang, D.M., 2022a. Effect of infection with Aeromonas hydrophila on antioxidant capacity, inflammation response, and apoptosis proteins in Chinese mitten crab (Eriocheir sinensis). Comp. Biochem. Physiol. C Toxicol. Pharmacol. 252, 109220.

Zheng, S., Xue, T., Wang, B., Guo, H., Liu, Q., 2022b. Chinese medicine in the treatment of ulcerative colitis: the mechanisms of signaling pathway regulations. Am. J. Chin. Med. 50, 1781–1798.

Zhou, H., Zhao, W., Zheng, Z., Aweya, J.J., Zhang, Y., Zhu, J., Zhao, Y., Chen, X., Yao, D., 2021. The notch receptor-ligand Delta is involved in the immune response of Penaeus vannamei. Dev. Comp. Immunol. 125, 104147.

Zhou, K., Qin, Y., Song, Y., Zhao, K., Pan, W., Nan, X., Wang, Y., Wang, Q., Li, W., 2022. A novel Ig domain-containing C-type lectin triggers the intestine-Hemocyte Axis to regulate antibacterial immunity in crab. J. Immunol. 208, 2343–2362.